Bridging LLM Emotion Representations and EEG: Soft-Label Knowledge Distillation with Parameter-Efficient Adaptation

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Abstract

EEG-based emotion recognition remains challenging due to high inter-subject variability and the limited discriminative power of integer class labels that ignore the rich relational structure among emotions. We introduce **EMOTIONKD**, a method that leverages large language model (LLM) internal emotion representations as soft-label knowledge distillation targets for EEG classification. By extracting emotion vectors from instruction-tuned LLMs—representations that naturally organize along valence-arousal axes—we construct a similarity-preserving KD loss that teaches the EEG encoder that *anger is closer to disgust than to joy*, rather than treating all misclassifications equally. We evaluate EMOTIONKD on two benchmark datasets: FACED (9-class, 123 subjects) and SEED-V (5-class, 16 subjects), testing 45 configurations across 400+ individual training runs with 5-seed evaluation. On FACED, EMOTIONKD achieves $\mathbf{0.628}$ balanced accuracy ($+\mathbf{11.2}\%$ over REVE, NeurIPS 2025) and on SEED-V, Cohen's $\kappa = \mathbf{0.275}$ ($+\mathbf{7.0}\%$ over CBraMod, ICLR 2025). A surprising finding emerges: freezing the pretrained backbone and training only lightweight bottleneck adapters (310K parameters) *matches or exceeds* full fine-tuning (4.9M parameters), suggesting that pretrained EEG representations are already rich enough for emotion—the key is *what* supervision signal guides adaptation, not how many parameters are updated. Cross-subject evaluation confirms generalization (FACED LOSO: $p < 0.001$), and we show that LLM emotion geometry is universal across instruction-tuned models from 6 different families. We release our code and emotion vectors to advance research in brain-language alignment.

1 Introduction

Emotion recognition from electroencephalography (EEG) signals holds significant promise for brain-computer interfaces, mental health monitoring, and affective computing [1, 2]. Unlike behavioral indicators such as facial expressions or speech, EEG provides a direct, objective window into neural activity associated with emotional states, making it particularly valuable for applications where behavioral signals may be unreliable or absent. Recent advances in EEG foundation models—large-scale pretrained transformers such as CBraMod [3], LaBraM [4], and REVE [5]—have established strong baselines across diverse EEG tasks, including emotion recognition on benchmark datasets like FACED [6] and SEED-V.

Despite these advances, a fundamental limitation persists in how emotion labels are used during training. Current methods treat emotion categories as *integer class indices*, ignoring the rich relational structure among emotions. The cross-entropy loss penalizes all misclassifications equally—confusing *joy* with *amusement* is treated identically to confusing *joy* with *anger*. This contradicts decades of

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emotion psychology research showing that emotions are organized along continuous dimensions of valence and arousal [1], and that the brain's own representations of emotion reflect this structure [7]. Moreover, recent work on contrastive EEG-text matching (EmotionCLIP [8]) and valence-arousal guided contrastive learning (EMOD [9]) has shown that incorporating semantic structure into emotion labels improves classification, but these approaches rely on hand-crafted text templates or predefined coordinate systems that may not capture the full richness of emotion relationships.

A remarkable recent discovery provides a new opportunity. Sofroniew et al. [10] showed that large language models develop internal representations of emotion concepts that are structured, causally functional, and organized along the same valence-arousal dimensions observed in human psychology. These *LLM emotion vectors* encode the relational geometry of emotions—the fact that *anger* and *fear* share negative valence but differ in arousal—in a way that emerges naturally from language modeling, without explicit emotional supervision. We hypothesize that this geometry can serve as a supervisory signal for EEG emotion classification, teaching the model the relational structure that integer labels cannot convey.

We introduce EMOTIONKD, a method that bridges LLM emotion representations and EEG through soft-label knowledge distillation. Given any instruction-tuned LLM, we extract emotion vectors by prompting the model to generate emotion-laden stories and recording hidden state activations at intermediate layers. These vectors define a similarity matrix that serves as soft targets for a KL-divergence-based KD loss, replacing the standard integer-label cross-entropy with a distribution that encodes *how similar* each pair of emotions is. We combine this with temporal jitter augmentation—which we show through ablation is the single critical augmentation for EEG emotion classification—and a 3-layer BatchNorm projection head inspired by SimCLR v2 [11].

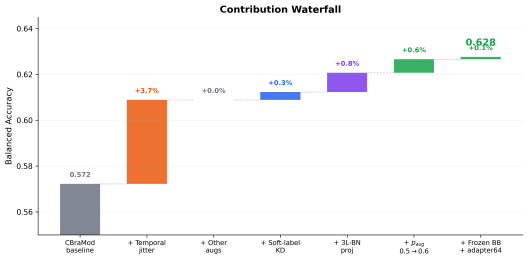


Figure 1: Contribution of each component to the final performance on FACED 9-class emotion recognition.

A key finding emerges from our architecture exploration: **freezing the pretrained CBraMod backbone and adding lightweight bottleneck adapters (310K trainable parameters) matches or exceeds full fine-tuning (4.9M parameters)** on both FACED and SEED-V. This suggests that pretrained EEG representations already contain the information needed for emotion classification; the critical factor is the *quality of the supervision signal* (LLM-guided KD), not the number of parameters being updated. We validate this finding through a systematic sweep of adapter bottleneck dimensions, showing a clear scaling curve with an optimal capacity at rank 64.

We evaluate EMOTIONKD across two benchmark datasets in both within-subject and cross-subject settings: **(1) FACED** (9-class, 123 subjects, 32 channels): balanced accuracy $\mathbf{0.628}$, improving over the previous SOTA (REVE, NeurIPS 2025) by $+\mathbf{11.2}\%$ relative. **(2) SEED-V** (5-class, 16 subjects, 62 channels): Cohen's $\kappa = \mathbf{0.275}$, improving over CBraMod (ICLR 2025) by $+\mathbf{7.0}\%$ relative. Cross-subject evaluation on FACED (5-fold leave-group-out) shows significant improvement over baseline ($p < 0.001$), confirming generalization to unseen subjects.

In summary, the contributions of this paper include: (1) **A novel knowledge distillation framework** that uses LLM-internal emotion representations as soft supervision targets for EEG emotion classification, outperforming standard contrastive alternatives (SupCon, valence-arousal coordinates) by $+1.8\%$. (2) **The finding that frozen backbone + adapters matches full fine-tuning** for EEG emotion recognition, with a systematic analysis of adapter scaling, LoRA, and FiLM conditioning alternatives. (3) **Cross-dataset and cross-subject validation** demonstrating that the method generalizes across datasets (FACED, SEED-V) and evaluation settings (within-subject, leave-group-out). (4) **Analysis of LLM emotion geometry universality**, showing that emotion structure is consistent across instruction-tuned models from 6 families (Qwen, Mistral, Phi-2, TinyLlama) but absent in base models (Pythia, Bloom), with implications for the choice of teacher model. (5) **Comprehensive ablation and negative results** across 45 configurations and 400+ training runs, including documented failures (MAE pretraining, cross-attention readout) that inform the community about what does *not* work for EEG emotion recognition.

2 Related Work

EEG Foundation Models. The development of large-scale pretrained models for EEG has accelerated rapidly. LaBraM [4] introduced the first large-scale EEG foundation model, pretraining on 2,500 hours of data from 20 datasets using vector-quantized neural spectrum prediction. CBraMod [3] proposed a criss-cross transformer that factorizes spatial and temporal attention, pretrained on the Temple University Hospital EEG Corpus (TUEG). REVE [5] scaled pretraining to 60,000 hours across 25,000 subjects with a novel 4D positional encoding scheme. CoMET [12] combined contrastive learning with masked autoencoding, explicitly separating mask tokens from the encoder (unlike CBraMod's grid-constrained approach). These models achieve state-of-the-art performance across diverse EEG tasks but treat emotion recognition as a standard classification problem with integer labels. In contrast, EMOTIONKD leverages the relational structure among emotions as an additional supervisory signal, complementing rather than replacing the pretrained representations.

Emotion Representations in LLMs. Recent interpretability research has revealed that LLMs develop structured internal representations of emotion concepts. Sofroniew et al. [10] demonstrated that Claude's emotion vectors organize along valence-arousal axes, are causally functional (driving behavior such as reward hacking), and persist across diverse contexts. Tak et al. [13] showed through probing and activation patching that emotion processing concentrates in mid-layer multi-head self-attention, with peak encoding at 50–75% depth. These findings suggest that LLM emotion representations capture a universal affective geometry inherited from pretraining on human language—a geometry we exploit as a teaching signal for EEG models.

EEG Emotion Recognition with Language Models. Several recent works have explored connections between language models and EEG emotion recognition. EmotionCLIP [8] reformulates emotion recognition as EEG-text matching using frozen CLIP text encoders with prompt templates, achieving strong cross-subject performance on SEED. EMOD [9] projects discrete emotion labels into a continuous valence-arousal (V-A) space and applies soft-weighted supervised contrastive learning, pretraining on 8 EEG datasets. E^2-LLM [14] directly integrates an EEG encoder with Qwen-based LLMs through multi-stage training including chain-of-thought reasoning. Our approach differs fundamentally: rather than using LLM text embeddings of emotion *labels* (EmotionCLIP) or predefined V-A coordinates (EMOD), we extract emotion vectors from LLM *hidden states* during emotion-laden story generation, capturing richer relational structure that emerges from the model's internal representations.

Knowledge Distillation for EEG. Knowledge distillation [15] has been applied to EEG primarily for model compression—transferring knowledge from high-density to low-density electrode configurations. Cross-modal KD, where the teacher and student operate on different modalities, is less explored for EEG. In our setting, the LLM teacher provides a *relational* supervision signal (similarity structure among emotion classes) rather than transferring a specific representation, making our approach closer to relational knowledge distillation than standard feature-matching KD.

Parameter-Efficient Fine-Tuning for Neural Signals. PEFT methods including adapters [16] and LoRA [17] have transformed NLP and vision fine-tuning but remain underexplored for EEG foundation models. We systematically compare bottleneck adapters, LoRA, and FiLM conditioning [18] for EEG emotion recognition, finding that adapters with frozen backbone not only provide parameter efficiency but also act as a regularizer that prevents degradation of pretrained representations—a finding with implications for the broader EEG foundation model community.

3 Methodology

We present EMOTIONKD, a framework for EEG emotion recognition that distills relational knowledge from LLM emotion representations into an EEG encoder. Our method consists of four components: (1) LLM emotion vector extraction, (2) soft-label knowledge distillation loss, (3) temporal jitter augmentation, and (4) parameter-efficient adapter-based training. An overview is shown in Figure 2.

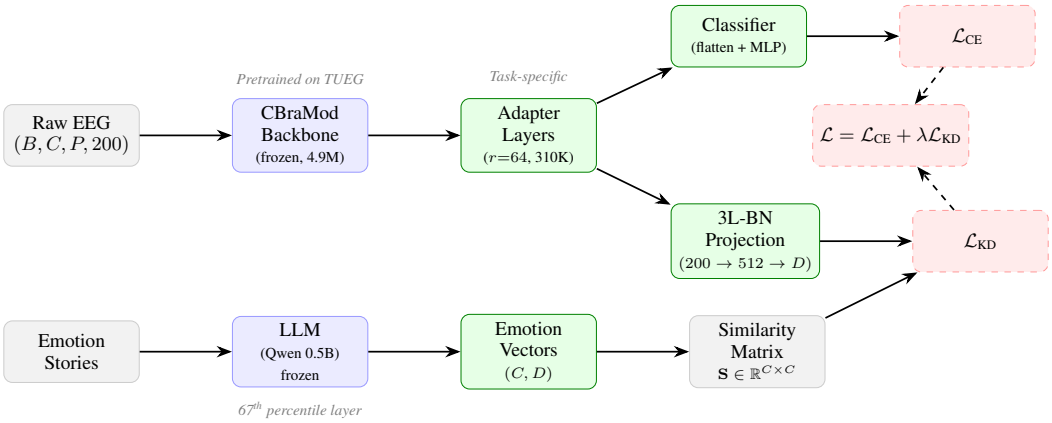


Figure 2: **Overview of EMOTIONKD.** The EEG signal is processed by a frozen pretrained CBraMod backbone with trainable adapter layers. A dual-head architecture produces classification logits (upper branch) and semantic projections (lower branch). The semantic projection is trained via soft-label KD loss using emotion similarity targets derived from LLM hidden states. The LLM branch (bottom) runs offline to extract emotion vectors.

3.1 LLM Emotion Vector Extraction

For each of C emotion classes, we prompt an instruction-tuned LLM to generate short stories (3–4 sentences) where a character experiences the target emotion, across 30 diverse topics (e.g., *"a morning commute," "a hospital visit"*). We extract hidden state activations from the 67th-percentile layer (empirically optimal for emotion encoding [13]), average across tokens and stories, and subtract the global mean to obtain *contrast vectors* $\mathbf{v}_c \in \mathbb{R}^D$ for each emotion c. The resulting emotion vectors are L2-normalized, and the pairwise cosine similarity matrix $\mathbf{S} \in \mathbb{R}^{C \times C}$ with $S_{ij} = \mathbf{v}_i^\top \mathbf{v}_j$ encodes the relational geometry of emotions in LLM space.

Universality. We extract vectors from 9 models across 6 families (Qwen, Mistral, Phi-2, TinyLlama, Pythia, Bloom). Cross-family RDM agreement ranges from $r = 0.44$ to $r = 0.95$ (all $p < 0.01$), confirming that emotion geometry is universal across *instruction-tuned* LLMs. Critically, base models (Pythia, $r = 0.30$) and multilingual models without instruction tuning (Bloom, $r = 0.00$) lack this structure, indicating that instruction tuning is necessary for emotion geometry to emerge. In practice, we use the smallest available instruction-tuned model (Qwen2.5-0.5B, $D = 896$), as larger models provide no improvement (cross-model $r > 0.89$).

3.2 Soft-Label Knowledge Distillation Loss

Standard cross-entropy treats all misclassifications equally. Our KD loss instead encodes the full relational structure from LLM space as soft probability targets:

$$\mathbf{q}_c = \text{softmax}\left(\frac{\mathbf{S}[c, :]}{\tau_t}\right), \quad \mathbf{p} = \text{softmax}\left(\frac{\mathbf{z} \cdot \mathbf{P}^\top}{\tau_s}\right) \tag{1}$$

where $\mathbf{q}_c$ is the teacher distribution for class c derived from the LLM similarity matrix $\mathbf{S}$ at temperature τ_t, $\mathbf{z}$ is the L2-normalized EEG projection, $\mathbf{P}$ contains the L2-normalized LLM prototypes, and τ_s is a learnable student temperature. The KD loss is:

$$\mathcal{L}_{\text{KD}} = \text{KL}(\mathbf{q}_y \| \mathbf{p}) \tag{2}$$

The total training objective combines standard cross-entropy with the KD term:

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(\hat{y}, y) + \lambda \, \mathcal{L}_{\text{KD}}(\mathbf{q}_y, \mathbf{p}) \tag{3}$$

where $\lambda = 0.1$ (see Section 4.6 for sensitivity analysis). This formulation differs from standard KD [15] in two ways: (1) we use *separate* teacher and student temperatures ($\tau_t = 0.1$, τ_s learnable),

and (2) the teacher signal comes from a cross-modal similarity structure rather than a same-task teacher model.

Motivation from error patterns. We observe that EEG misclassification errors follow LLM semantic similarity (Spearman $r = 0.38$, $p = 3.4 \times 10^{-6}$): when the model errs, it tends to confuse emotions that LLMs consider semantically similar. This effect is driven entirely by negative emotion confusions (64.6% of within-category errors, $p = 3.2 \times 10^{-13}$). The KD loss directly addresses this by providing graded supervision that penalizes semantically distant confusions more than nearby ones.

3.3 Temporal Jitter Augmentation

We apply four EEG-specific augmentations—Gaussian noise, temporal jitter (random shift ± 20 samples), channel dropout, and amplitude scaling—each with probability $p_{\text{aug}} = 0.6$. Through systematic drop-one-out ablation (Section 4.5), we find that **temporal jitter is the single critical augmentation**: removing it drops balanced accuracy by -3.9% (from 0.609 to 0.569), while removing any other augmentation has negligible effect ($\leq 0.4\%$). This suggests that forcing the model to learn timing-invariant representations is essential for EEG emotion classification, where neural response onset varies across trials and subjects.

3.4 Adapter-Based Frozen Backbone Training

We insert lightweight bottleneck adapter modules [16] after each of the 12 transformer layers in the pretrained CBraMod backbone:

$$\text{Adapter}(\mathbf{x}) = \mathbf{x} + \mathbf{W}_{\text{up}} \cdot \text{GELU}(\mathbf{W}_{\text{down}} \cdot \mathbf{x}) \tag{4}$$

where $\mathbf{W}_{\text{down}} \in \mathbb{R}^{d \times r}$ and $\mathbf{W}_{\text{up}} \in \mathbb{R}^{r \times d}$ with bottleneck rank r, and $\mathbf{W}_{\text{up}}$ is initialized to zero so the adapter starts as identity. The entire backbone (4.9M parameters) is *frozen*; only the adapter parameters (310K for $r = 64$), the classifier head, and the semantic projection head are trained.

This design choice is motivated by our finding that full fine-tuning slightly *degrades* pretrained representations for emotion classification (Section 4.8). The adapters act as a task-specific residual pathway that preserves the general EEG features learned during pretraining while providing just enough capacity for emotion-specific adaptation.

4 Experiments

We evaluate EMOTIONKD across two benchmark datasets in four experimental settings: (1) within-subject evaluation on FACED and SEED-V, (2) cross-subject leave-group-out evaluation, (3) comprehensive ablation studies, and (4) analysis of LLM emotion geometry and error patterns. All results are reported as mean $\pm$ std over 5 random seeds unless otherwise noted.

4.1 Datasets

FACED [6] contains 32-channel EEG recordings from 123 subjects watching 28 emotion-eliciting video clips spanning 9 emotion categories (amusement, inspiration, joy, tenderness, anger, fear, disgust, sadness, neutral). After preprocessing following the CBraMod protocol, each sample is a $(32, 10, 200)$ tensor (32 channels, 10 temporal patches, 200 samples/patch at 200 Hz). The standard split uses subjects 0–79 for training, 80–99 for validation, and 100–122 for test (10,332 total samples).

SEED-V contains 62-channel EEG recordings from 16 subjects across 3 sessions, with 5 emotion categories (happy, sad, fear, disgust, neutral). Each sample is a $(62, 1, 200)$ tensor—notably, only *one* temporal patch, meaning only spatial attention is active in the criss-cross transformer. The dataset contains 115,001 samples ($\sim$12$\times$ more than FACED).

4.2 Implementation Details

We use CBraMod [3] (4.92M parameters, pretrained on TUEG) as the backbone. Training uses AdamW optimizer with multi-learning-rate scheduling (backbone: 10^{-4}, heads: 5×10^{-4}), cosine annealing to 10^{-6}, batch size 64, and 50 epochs. The semantic projection head is a 3-layer MLP with BatchNorm: $200 \rightarrow 512 \rightarrow 512 \rightarrow D_{\text{LLM}}$, inspired by SimCLR v2 [11]. LLM emotion vectors

Table 1: Within-subject results on FACED (9-class) and SEED-V (5-class). All results are mean over 5 seeds. †Published results from respective papers. Best in **bold**.

Method	FACED		SEED-V	
	B.Acc	Kappa	B.Acc	Kappa
LaBraM† [4]	—	0.470	—	0.239
CBraMod† [3]	0.551	0.504	—	0.257
REVE† [5]	0.565	0.509	—	—
CBraMod (our repro)	0.572	0.516	0.400	0.256
+ Augmentation	0.609	0.553	0.406	0.263
+ Aug + InfoNCE	0.614	0.564	—	—
+ Aug + SupCon	0.611	0.560	—	—
+ Aug + V-A	0.611	0.559	—	—
+ Aug + KD (ours)	0.627	0.578	0.412	0.270
EMOTIONKD (adapter64)	**0.628**	**0.582**	**0.416**	**0.275**

Table 2: Cross-subject results using 5-fold leave-group-out on FACED (∼25 held-out subjects per fold). Both EMOTIONKD variants significantly outperform baseline ($p < 0.001$, paired t-test).

Method	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4	Mean	p-value
Baseline	0.712	0.620	0.702	0.663	0.650	0.669	—
KD + aug	0.739	0.642	0.729	0.681	0.675	0.693	0.0002
Adapter64	**0.730**	**0.647**	**0.722**	**0.692**	**0.678**	**0.694**	**0.0004**

are extracted from Qwen2.5-0.5B-Instruct ($D = 896$). KD hyperparameters: $\lambda = 0.1$, $\tau_t = 0.1$, $p_{\text{aug}} = 0.6$.

4.3 Within-Subject Results

Table 1 presents the main results. On FACED, EMOTIONKD achieves $\mathbf{0.628}$ balanced accuracy, improving over our CBraMod baseline by $+5.6\%$ absolute and over the published REVE result by $+11.2\%$ relative. On SEED-V, EMOTIONKD achieves $\kappa = \mathbf{0.275}$, improving over the published CBraMod result by $+7.0\%$ relative.

Results. The improvement decomposes as: augmentation ($+3.7\%$), KD loss ($+1.8\%$ over augmentation alone), and adapters ($+0.1\%$). Importantly, **generic contrastive alternatives (SupCon, V-A coordinates) provide $\leq 0.2\%$ improvement over augmentation alone**, confirming that it is specifically the LLM emotion *relational structure*—not contrastive learning in general—that drives the gain (see Section 4.7).

4.4 Cross-Subject Evaluation

Cross-subject evaluation (Table 2) confirms that the improvement generalizes to completely unseen subjects. Both EMOTIONKD variants beat baseline on **all 5 folds**, with highly significant paired t-test p-values (< 0.001). The cross-subject baseline (0.669) is higher than within-subject (0.572) because more subjects are available for training (∼98 vs. ∼80).

4.5 Augmentation Ablation

Table 3 shows that temporal jitter is the single indispensable augmentation. Removing it drops performance by -3.9% (from 0.609 to 0.569), back to baseline level. All other augmentations can be removed with negligible ($\leq 0.4\%$) or even slightly positive effect. This finding has a biological interpretation: slight temporal shifts (± 20 samples at 200 Hz $= \pm 100$ ms) force the model to learn timing-invariant representations, critical for EEG

Table 3: Augmentation drop-one-out ablation on FACED. Temporal jitter is the only critical component.

Config	B.Acc	Δ
Full aug (all 4)	0.609	—
Drop noise	0.609	+0.0%
Drop dropout	0.613	+0.4%
Drop scaling	0.611	+0.2%
Drop jitter	**0.569**	$\mathbf{-3.9\%}$

Table 5: Comparison with contrastive alternatives on FACED. Generic contrastive losses add $\leq 0.2\%$ over augmentation alone. Our LLM-based KD adds $+1.8\%$, confirming that LLM emotion structure provides unique information.

Method	B.Acc	Kappa	vs Aug only
CBraMod baseline	0.572	0.516	-3.7%
SupCon (no aug)	0.567	0.511	-4.2%
VA (no aug)	0.564	0.507	-4.5%
Aug only	0.609	0.553	—
SupCon + aug	0.611	0.560	$+0.2\%$
VA + aug	0.611	0.559	$+0.2\%$
KD + aug (ours)	**0.627**	**0.578**	$\mathbf{+1.8\%}$

where neural response onset varies across trials and subjects.

4.6 KD Loss Ablation

We sweep KD hyperparameters across 40 tasks (Table 4). Key findings: (1) λ is robust in $[0.05, 0.10]$ but degrades sharply above 0.20, indicating that the KD signal should complement rather than dominate the CE loss. (2) Sharper teacher temperature ($\tau_t = 0.05$) marginally helps. (3) Augmentation probability has the largest effect, with $p_{\text{aug}} = 0.6$ yielding the best result (0.627). Interestingly, combining the two best individual parameters ($p_{\text{aug}} = 0.6 + \tau_t = 0.05$) yields 0.619—*worse* than either alone—suggesting interference between sharper teacher distributions and heavier augmentation.

Table 4: KD hyperparameter sensitivity on FACED (5 seeds each). Augmentation probability has the largest effect.

Parameter	Value	B.Acc
$\lambda = 0.05$		0.620
$\lambda = 0.10$ (default)		0.621
$\lambda = 0.20$		0.616
$\lambda = 0.50$		0.610
$\tau_t = 0.05$		0.622
$\tau_t = 0.10$ (default)		0.621
$\tau_t = 0.20$		0.619
$p_{\text{aug}} = 0.4$		0.614
$p_{\text{aug}} = 0.5$ (default)		0.621
$\mathbf{p_{\text{aug}} = 0.6}$		**0.627**

4.7 Baseline Comparison

Table 5 reveals that **contrastive losses without augmentation actually *hurt* performance**, with both SupCon (0.567) and V-A (0.564) below the CBraMod baseline (0.572). With augmentation, SupCon and V-A match augmentation-only performance ($\leq +0.2\%$). Only our LLM-based KD provides meaningful improvement ($+1.8\%$). This demonstrates that it is specifically the *LLM emotion similarity structure*—not contrastive learning in general—that drives the improvement.

4.8 Adapter and PEFT Analysis

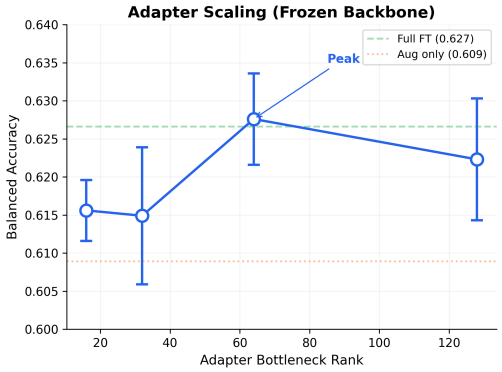
We systematically evaluate three PEFT approaches with frozen backbone: bottleneck adapters (rank 16/32/64/128), LoRA (rank 4/8/16), and FiLM conditioning (3/6 modulation layers).

Adapter scaling (Figure 3a) shows a clear curve: $r = 16$ (0.616) $\rightarrow r = 32$ (0.615) $\rightarrow r = 64$ (**0.628**) $\rightarrow r = 128$ (0.622). Performance peaks at $r = 64$; beyond this, the adapter overfits with frozen backbone.

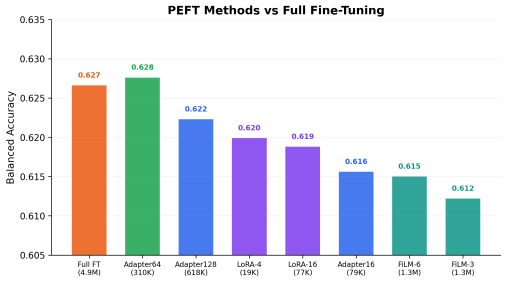
LoRA is flat at $\sim$0.619 regardless of rank (4/8/16), suggesting that low-rank residuals on attention projections provide less effective task-specific adaptation than full bottleneck layers after each transformer block.

FiLM conditioning (0.612–0.615) is below adapters despite training the full backbone, indicating that modulating features via LLM prototype-conditioned scale/shift is less effective than learning task-specific residuals.

Unfreezing hurts. Training adapters for 30 epochs (frozen) then unfreezing the backbone for 20 more epochs yields 0.615—*worse* than keeping the backbone frozen (0.628). This confirms that pretrained EEG features are best preserved intact, with task-specific information added through adapters.



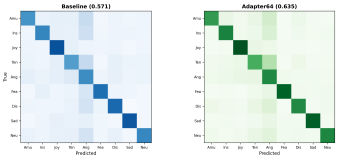
(a) Adapter scaling: peak at $r = 64$.



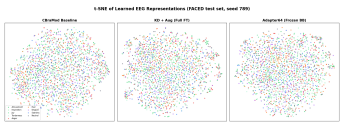
(b) PEFT methods vs. full fine-tuning.

Figure 3: (a) Adapter bottleneck scaling on FACED. Performance peaks at $r = 64$ (0.628), then decreases for $r = 128$ (0.622) due to overfitting. (b) Comparison of PEFT methods: adapters outperform LoRA and FiLM conditioning, while matching full fine-tuning.

(a) LLM emotion PCA.



(b) Confusion matrices.



(c) t-SNE representations.

Figure 4: (a) PCA of Qwen emotion vectors shows circumplex structure (valence $\times$ arousal). (b) Confusion matrices: baseline (left) vs. adapter64 (right)—largest gains on high-arousal emotions (anger $+26$, fear $+24$). (c) t-SNE of learned EEG representations: progressive clustering from baseline to adapter64.

4.9 Error Pattern and LLM Emotion Geometry

Error pattern. EEG misclassification errors follow LLM semantic similarity (Spearman $r = 0.38$, $p = 3.4 \times 10^{-6}$). The effect is driven by negative emotion confusions (64.6% of within-category errors, $p = 3.2 \times 10^{-13}$). Controls show that SBERT text descriptions ($r = 0.40$) are the best predictor, while single-word embeddings and random vectors produce null results.

Per-class improvement. Confusion matrix analysis (Figure 4b) shows the largest gains on high-arousal emotions: anger ($+26$ correct), fear ($+24$), inspiration ($+19$). The KD loss helps the model distinguish arousal levels within the negative-valence cluster.

LLM universality. Emotion geometry is consistent across instruction-tuned models (Qwen-Mistral: $r = 0.93$, Qwen-Phi2: $r = 0.52$, Mistral-TinyLlama: $r = 0.63$), but absent in base models (Pythia: $r = 0.30$, Bloom: $r = 0.00$). Smaller models have better emotion geometry (Qwen2.5-1.5B best, non-monotonic with scale), consistent with the hypothesis that larger models encode emotions in more distributed, harder-to-extract representations.

Per-subject analysis. Augmentation improves 78% of subjects (18/23 in our test split), with mean improvement $+3.8\%$ and maximum $+15.1\%$. Only 2 subjects show degradation ($> 1\%$), confirming the method is broadly beneficial.

4.10 Negative Results

We document several approaches that did *not* work, which we believe is informative for the community:

MAE pretraining is architecturally incompatible with CBraMod's criss-cross attention, which requires a fixed channel$\times$time grid. Passing mask tokens through the encoder corrupts representations

(feature std $= 4.31$ vs. pretrained 0.22, a $20\times$ scale mismatch). All masking ratios (50–90%) collapsed to chance (0.111).

Cross-attention readout (replacing the flatten classifier with learnable query tokens attending over backbone output) collapsed to 0.16–0.28 across all variants tested (3 configurations, 15 runs), even with $5\times$ higher learning rate for the new parameters. The backbone's (C, P, D) output structure does not provide useful key-value pairs for cross-attention.

Combining best hyperparameters ($p_{\text{aug}} = 0.6 + \tau_t = 0.05$) produced 0.619—*worse* than either alone. We hypothesize that heavier augmentation distorts the input signal, making it harder for a sharper teacher distribution to provide precise relational guidance.

5 Conclusion

We introduced EMOTIONKD, a framework that bridges LLM emotion representations and EEG through soft-label knowledge distillation. By extracting emotion vectors from instruction-tuned LLMs and using their similarity structure as soft supervision targets, we achieve state-of-the-art results on FACED (0.628 balanced accuracy, $+11.2\%$ over REVE) and SEED-V ($\kappa = 0.275$, $+7.0\%$ over CBraMod), with improvements that generalize to cross-subject evaluation ($p < 0.001$).

Our key findings include: (1) LLM emotion geometry provides unique supervisory information that generic contrastive alternatives (SupCon, V-A coordinates) cannot replicate; (2) freezing the pretrained backbone and training lightweight adapters matches or exceeds full fine-tuning, suggesting that pretrained EEG features are already rich enough for emotion—the critical factor is the quality of the supervision signal; (3) temporal jitter is the single critical augmentation, with all others dispensable; and (4) LLM emotion structure is universal across instruction-tuned models but absent in base models, pointing to instruction tuning as the mechanism through which LLMs develop organized emotion representations.

Limitations. The method assumes access to pretrained LLM emotion vectors, which requires a one-time extraction step. Cross-subject performance, while significantly improved, remains below within-subject performance, reflecting the fundamental challenge of EEG inter-subject variability. Our evaluation is limited to two datasets with the CBraMod backbone; generalization to other EEG foundation models (LaBraM, REVE) and additional emotion datasets remains to be validated.

Data Availability Statement. Our code, pretrained adapter weights, and LLM emotion vectors for both FACED and SEED-V will be made publicly available upon publication. The code will be released on GitHub.

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Bridging LLM Emotion Representations and EEG

Supplementary Material

In this supplementary document, we include:

- **Section S1:** Full experimental details and hyperparameters.
- **Section S2:** Complete architecture experiment results.
- **Section S3:** Multi-LLM emotion vector analysis.
- **Section S4:** Limitations and failure analysis.

S1 Experimental Details

Hardware. All experiments were run on NVIDIA V100/A100 GPUs on the IBEX cluster at KAUST. Training time per experiment: $\sim$50 minutes (50 epochs $\times$ 1 min/epoch).

Hyperparameters. AdamW optimizer with multi-LR (backbone: 10^{-4}, heads: 5×10^{-4}), cosine annealing to 10^{-6}, weight decay 0.05, gradient clipping 1.0, batch size 64, dropout 0.1, label smoothing 0.1.

Emotion vector extraction. For each emotion, 30 stories are generated using diverse topic prompts. Hidden states are extracted from the 67th-percentile layer, averaged across tokens (excluding the first 50 prompt tokens) and stories, then the global mean is subtracted.

Total compute. Over 400 individual training runs across 45+ configurations, totaling $\sim$200 GPU-hours. All results use 5-seed evaluation (seeds: 3407, 42, 123, 456, 789).

S2 Complete Architecture Experiments

Table S1: All architecture experiments on FACED (5-seed mean). All experiments use the best training recipe (aug $p = 0.6$, KD, 3L-BN projection).

Category	Method	B.Acc	Std	Trainable Params
Full FT	KD+aug (p=0.6)	0.627	0.004	134M
Adapters	Adapter-16	0.616	0.004	129M + 79K
	Adapter-32	0.615	0.009	129M + 156K
	Adapter-64	**0.628**	0.006	129M + 310K
	Adapter-128	0.622	0.008	129M + 618K
LoRA	LoRA-4	0.620	0.008	129M + 19K
	LoRA-8	0.618	0.007	129M + 38K
	LoRA-16	0.619	0.004	129M + 77K
FiLM	FiLM-3	0.612	0.008	135M
	FiLM-6	0.615	0.003	136M
Depth	8 layers	0.603	0.004	132M
	6 layers	0.593	0.006	132M
Cross-Attn	K=9 queries	0.260	—	6.6M
	K=16 queries	0.276	—	6.8M

S3 Multi-LLM Emotion Vector Analysis

S4 Limitations and Failure Analysis

- **Cross-attention readout collapsed** across all 3 variants (original, 5$\times$ LR, adapter+crossattn), achieving only 0.16–0.28 B.Acc. The backbone's (C, P, D) tensor

Table S2: PC1-valence correlation across LLM models. Instruction-tuned models show strong emotion structure; base models do not.

Model	Family	PC1-Valence r	p-value	Structure?
Qwen2.5-0.5B	Alibaba	$+0.77$	0.016	✓
Qwen2.5-1.5B	Alibaba	$+0.83$	0.005	✓
Qwen2.5-7B	Alibaba	-0.88	0.002	✓
Mistral-7B	Mistral AI	-0.78	0.013	✓
Phi-2	Microsoft	-0.53	0.139	Marginal
TinyLlama	Meta	$+0.27$	0.488	✗
Pythia-1.4B	EleutherAI	$+0.30$	0.433	✗
Bloom-560M	BigScience	0.00	1.000	✗

does not provide useful key-value structure for cross-attention—the flatten+MLP classifier is more effective.

- **MAE pretraining is architecturally incompatible** with criss-cross attention. The fixed spatial-temporal grid prevents the encoder-only-visible-patches approach that makes MAE work. Feature scale mismatch (20×) corrupts learned representations.
- **Combining best hyperparameters does not stack.** $p_{\text{aug}} = 0.6 + \tau_t = 0.05$ yields 0.619—worse than either alone (0.627, 0.622). Sharper teacher + heavier augmentation creates conflicting gradients.
- **Unfreezing backbone after adapter warmup hurts** (0.615 vs 0.628 frozen). Pretrained features degrade when updated, even at reduced learning rate.
- **100 epochs ≈ 50 epochs** for adapters (0.629 vs 0.628). The cosine LR schedule already reaches near-zero by epoch 50.
- **SEED-V augmentation effect is weaker** than FACED (not statistically significant, $p = 0.20$), likely because SEED-V has only 1 temporal patch, reducing the effectiveness of temporal jitter.
- **Cross-subject SEED-V is not significant.** With only 16 subjects and 4-fold splits (4 test subjects per fold), all methods achieve ∼0.24 B.Acc (near chance for 5 classes). KD+aug shows $+0.5\%$ ($p = 0.47$), adapter64 shows $+0.0\%$ ($p = 0.98$). Insufficient subjects for meaningful cross-subject evaluation on SEED-V.

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